

STANTON, MN, USA

WMO: 722003

Lat: 44.476N

Lon: 93.016W

Elev: 920

StdP: 14.21

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54930

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-7.8	-20.0	1.9	-11.7	-15.4	2.5	-7.2	26.5	18.7	24.2	17.7	4.5	210	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	89.6	73.7	86.4	71.6	83.9	70.0	76.7	86.1	74.4	83.2	72.5	80.8	10.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	128.4	83.6	71.5	120.3	81.0	69.7	113.0	78.4	40.7	86.5	38.6	83.7	36.7	80.7	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.6	20.2	18.1	DB	-20.2	94.7	6.3	3.6	-24.8	97.3	-28.5	99.4	-32.0	101.4	-36.6	104.0	
			WB	-20.4	79.5	6.0	2.2	-24.7	81.1	-28.3	82.3	-31.6	83.6	-36.0	85.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	16.3	18.4	33.2	46.0	59.4	68.3	72.1	69.2	62.6	49.3	34.8	21.2
	DBStd	22.19	13.59	12.45	12.92	9.99	8.98	6.04	5.43	5.58	8.24	9.73	10.66	11.87
	HDD50	4169	1044	884	538	189	17	0	0	0	6	132	468	891
	HDD65	7557	1509	1304	987	571	224	32	7	21	141	498	907	1356
	CDD50	2731	0	0	18	70	308	550	685	594	384	111	11	0
	CDD65	644	0	0	1	2	50	132	227	150	70	12	0	0
	CDH74	5892	0	0	8	36	531	1122	2160	1277	641	116	1	0
	CDH80	1692	0	0	0	3	173	319	681	326	165	25	0	0
	WSAvg	7.6	8.3	8.3	8.4	9.7	8.7	7.0	5.6	4.9	6.4	7.7	8.7	7.8
	PrecAvg	34.3	0.9	0.7	2.1	3.0	4.1	5.1	4.4	5.0	3.5	2.4	1.7	1.1
Precipitation	PrecMax	41.1	3.0	1.8	3.8	5.8	7.6	8.6	8.4	10.3	11.2	6.1	5.2	3.1
	PrecMin	23.0	0.2	0.1	0.4	0.7	1.6	2.7	1.3	1.7	0.9	0.5	0.2	0.3
	PrecStd	4.5	0.7	0.5	0.9	1.4	1.8	1.7	2.0	2.0	2.4	1.5	1.3	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.1	54.2	73.7	79.1	91.1	92.4	93.3	90.2	89.3	83.6	68.0	50.3
		MCWB	37.9	48.7	61.1	58.3	67.3	74.2	78.5	75.3	72.0	63.8	56.8	46.0
	2%	DB	39.1	42.1	63.8	72.6	84.6	87.2	89.8	86.4	84.7	77.0	60.5	43.2
		MCWB	35.5	37.8	55.3	56.2	65.9	70.6	76.3	72.8	70.6	61.6	51.4	39.6
	5%	DB	36.5	38.4	57.3	68.1	80.0	83.6	86.4	83.7	81.1	71.8	54.9	39.2
		MCWB	34.1	34.8	50.5	53.1	63.2	68.0	73.4	71.4	67.9	59.5	47.4	36.4
	10%	DB	34.1	35.1	50.4	63.4	75.1	80.4	83.6	81.1	77.4	65.2	50.3	36.5
		MCWB	32.2	32.4	44.7	50.8	61.0	66.2	71.1	69.4	66.4	54.9	43.9	34.2
	0.4%	WB	38.8	48.4	62.9	61.4	72.2	77.2	80.9	77.4	75.6	70.1	58.0	47.9
		MCDB	42.5	54.5	72.3	71.6	85.3	87.6	88.8	87.1	85.3	77.2	65.8	50.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.1	38.6	56.9	58.6	68.9	73.5	77.8	75.2	72.4	64.8	52.9	40.0
		MCDB	38.8	41.8	61.6	69.5	79.8	82.7	87.4	83.6	80.5	74.4	58.9	42.6
	5%	WB	34.3	35.0	50.0	55.8	66.0	71.0	75.3	73.1	70.2	59.6	47.9	36.5
		MCDB	36.3	37.9	57.0	65.1	74.9	79.9	84.2	80.8	78.3	68.8	53.8	38.8
	10%	WB	32.3	32.7	45.0	52.5	63.5	68.9	73.1	71.3	67.8	56.3	44.3	34.4
		MCDB	33.9	34.9	50.4	62.0	72.3	77.1	81.6	78.8	75.9	64.3	49.8	36.3
Mean Daily Temperature Range	5% DB	MDBR	15.9	17.1	16.6	19.5	20.9	18.8	20.3	20.3	21.5	19.9	16.0	14.0
		MCDBR	15.3	16.7	24.3	28.4	26.3	22.9	21.6	21.9	24.4	26.8	22.9	15.3
		MCWBR	13.1	13.5	15.4	15.7	11.7	10.2	10.8	11.0	12.2	13.8	15.1	12.8
	5% WB	MCDBR	14.6	15.7	22.2	23.5	20.2	19.0	19.9	19.2	21.0	20.5	21.1	14.0
		MCWBR	12.7	13.2	14.9	14.7	10.2	10.8	11.3	10.6	12.0	12.8	14.9	11.9
Clear-Sky Solar Irradiance	taub		0.267	0.292	0.320	0.374	0.403	0.405	0.403	0.403	0.374	0.324	0.291	0.268
	taud		2.420	2.335	2.384	2.293	2.265	2.303	2.320	2.320	2.370	2.501	2.503	2.440
	Ebn at Noon		275	285	291	281	275	275	273	269	268	271	264	263
	Edh at Noon		23	30	33	39	42	41	39	38	33	25	21	20
All-Sky Solar Radiation	RadAvg		551	895	1185	1446	1687	1886	2001	1675	1315	844	541	391
	RadStd		65	97	114	157	136	153	75	115	95	112	53	37

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page